Data Analysis Internship

Task 1 Report

Basic Sales Data Analysis of a Retail Superstore

Analysis carried out on

THE SUPERSTORE DATASET

51,290 orders, 2011 to 2015

Prepared by

KOUAME KOFFI FIDÈLE

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Contents

1 Data and method

1.1 Dataset

Table 1. Description of the dataset

Item	Detail
Name	Superstore Sales Dataset, 2011 to 2015
Source	Kaggle, `vivek468/superstore-dataset-final`
Records	51,290 orders
Columns in file	24
Columns used	7, as specified by the task
Period covered	January 2011 to December 2015
Distinct products	3,788

The seven fields used are the order date, the product name, the category, the region, the quantity ordered, the sales value and the profit. Profit is a signed quantity: a negative value denotes a loss-making order, and Section 2 explains why such orders were retained.

1.2 Tools

The analysis was carried out in Python 3 using pandas and NumPy for data manipulation, Matplotlib and Seaborn for the figures, and openpyxl for the spreadsheet export. The complete analysis is reproducible from the accompanying notebook, `Task1_Sales_Analysis.ipynb`, which was executed end to end before submission.

1.3 Method

Each of the six questions is answered by grouping the cleaned table and aggregating within groups, which is the appropriate operation: the questions ask for totals within categories rather than for a predictive model.

Two quantities were computed beyond those the task requires, because they change how the requested answers should be read. The first is the profit margin,

$$\text{margin} = 100 \times \frac{\sum \text{profit}}{\sum \text{sales}}, \tag{1}$$

which distinguishes a group that sells a great deal from a group that earns a great deal. The second is each group's share of the total, so that a contribution is expressed relative to the whole rather than as an absolute figure whose magnitude is difficult to judge.

2 Data cleaning

Cleaning proceeded in five steps, in a deliberate order: the data were first diagnosed without modification, then the types were corrected, then duplicates were removed, then invalid records, and finally the outliers were examined without being removed. The order matters. Diagnosing after correction would report the wrong counts, and removing duplicates before parsing the dates would compare character strings rather than dates.

2.1 Two defects that a default pipeline would have concealed

The order date column contains two distinct formats. 31,223 records use the form `d-m-Y` with hyphens, and 20,067 use the form `d/m/Y` with slashes. A single call to `pd.to_datetime` parses one of the two and converts the other to a null value without warning. Those records are then removed by the validity filter, which on this file discards sixty per cent of the data silently. The notebook parses each format explicitly and asserts that no date remains unparsed, so that a failure of this kind becomes an error rather than a missing result.

Duplicate detection must use the complete record. Once the order identifier, the customer and the city have been removed, two genuinely distinct orders may appear identical: the same product, on the same day, in the same region, for the same amount. The reduced table shows 2,347 apparent duplicates, whereas the original twenty-four column record shows none. Removing duplicates from the reduced table would therefore have deleted 2,347 real orders and understated revenue.

2.2 A decision stated explicitly

Records with a negative profit were retained. A loss-making order is not a data error; it is a real and consequential event, and it constitutes the principal finding of this study. Removing such records as outliers would have inflated the total profit and concealed the problem entirely.

Large orders were likewise examined and retained. In retail the largest orders are ordinarily genuine bulk purchases, and trimming them would remove precisely the revenue that a sales analysis exists to explain.

3 Results

3.1 The six questions

Table 2. Answers to the six questions set by the task

Question	Answer
Total sales	12,642,502
Total profit	1,467,457
Overall profit margin	11.61%
Category with the highest sales	Technology, 37.5% of the total
Best-selling product	Staples, by units sold
Region with the highest sales	Central, 22% of the total
Month with the highest sales	December
Orders analysed	51,290
Distinct products	3,788

The expression *best-selling product* is ambiguous, since ranking by units sold and ranking by revenue yield different answers: an inexpensive product sold in quantity outranks an expensive one sold rarely. Both rankings are reported in the notebook rather than one being selected silently.

3.2 Sales by category

Technology leads with 37.5 per cent of sales. The three categories are however close in revenue, spanning 30.0 to 37.5 per cent, so the interesting difference between them is not revenue but margin, examined in Section 4.

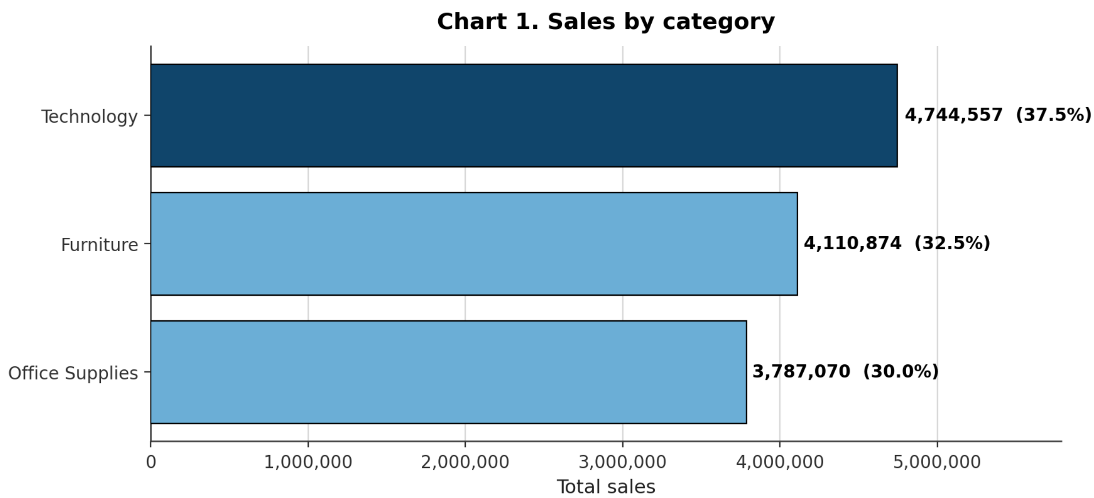

Figure 1. Total sales by product category. The leading category is shown in the darker tone. Values are annotated with the share of the total.

3.3 Sales over time

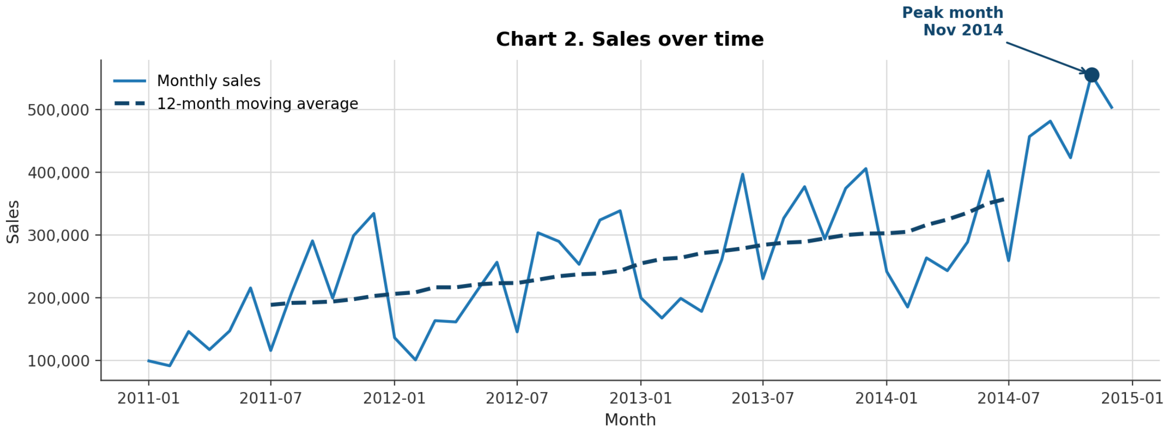

Figure 2. Monthly sales across the five-year period, with a centred twelve-month moving average. The window spans exactly one seasonal cycle, so the smoothed series isolates the underlying trend.

3.4 Seasonality

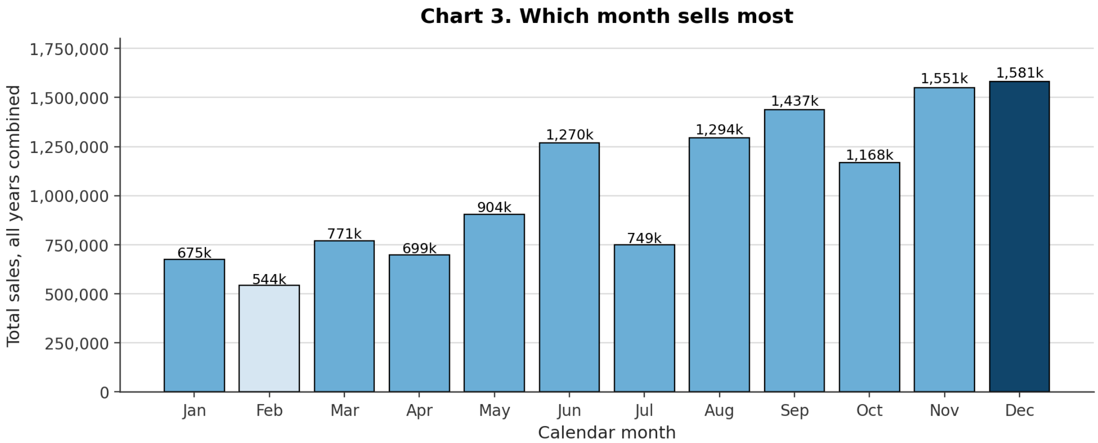

Figure 3. Total sales by calendar month, all years combined. This figure and the preceding one use the same data and answer different questions: the former shows when sales occurred, the latter which month sells most on average.

3.5 Leading products

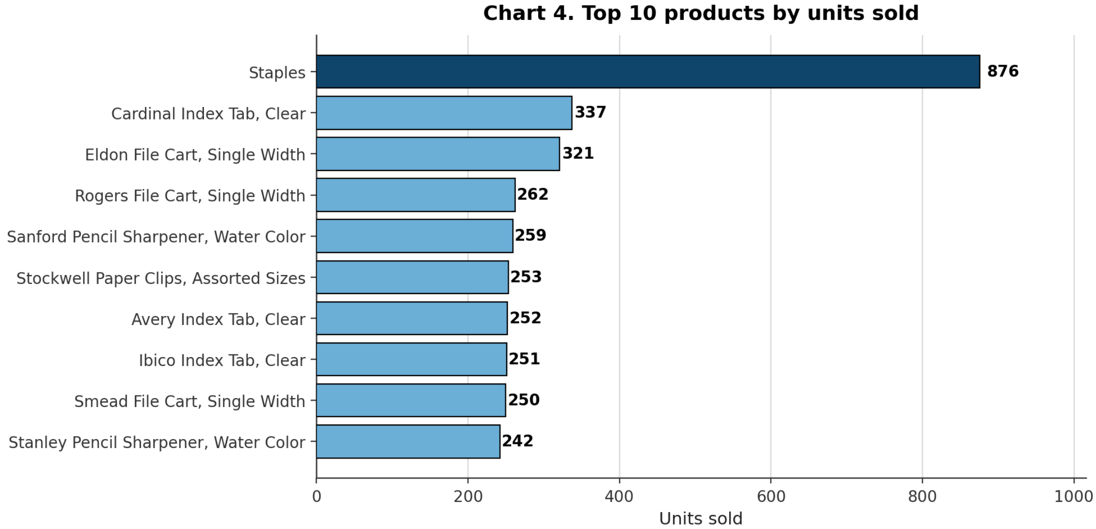

Figure 4. The ten products with the highest number of units sold. The catalogue contains 3,788 products; a complete chart would be unreadable and would answer nothing.

3.6 Regional performance

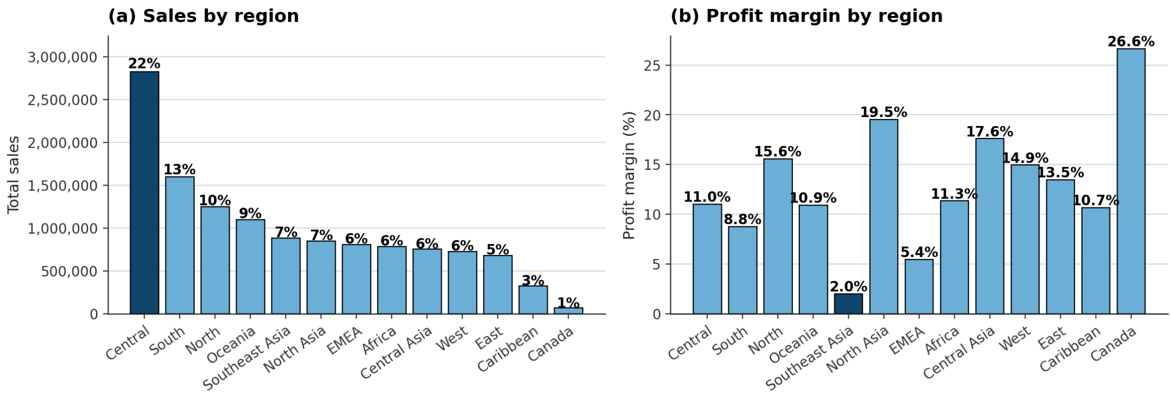

Figure 5. Sales and profit margin by region, on a common category order. The task asks which region sells most; the second panel answers the question a manager asks next, and the answers differ.

4 Insights

4.1 Technology leads on revenue, Furniture has the margin problem

Technology accounts for 37.5 per cent of sales, the largest share of the three categories. The ranking changes on profitability: Furniture returns the weakest margin of the three, roughly half that of Technology. Furniture therefore sells volume without converting it into profit, and it is the category to examine on pricing and discounting rather than on demand. A ranking based on sales alone would have presented it as healthy.

4.2 Nearly one order in four is loss making

12,544 of the 51,290 orders, that is 24.5 per cent, return a negative profit. This is the largest finding in the dataset and it is invisible in any total, because the profitable orders offset the loss-making ones: the reported profit of 1.47 million is a net figure that conceals a substantial gross loss.

It is also the reason for which such records were deliberately retained during cleaning. Treating them as outliers would have inflated the total profit and suppressed the finding.

4.3 Sales are strongly seasonal

December is the strongest month and February the weakest, the former selling approximately two and a half times the latter. A ratio of that magnitude is not noise. It justifies planning inventory, staffing and cash flow around the seasonal profile rather than against a flat annual average: planning to the mean guarantees a shortage in December and an excess in February.

4.4 Revenue is distributed across regions, profitability is not

Central leads on sales with 22 per cent of the total, whereas Southeast Asia converts its revenue at a margin of 2.0 per cent against 26.6 per cent in Canada. A region may be busy and barely profitable. This is the reason for which Figure 5 presents sales and margin side by side: a ranking based on sales alone would have presented Southeast Asia as a success on the strength of its seven per cent revenue share.

4.5 Revenue is concentrated in a small part of the catalogue

The store sells 3,788 distinct products, and revenue is concentrated in a small minority of them. The long tail carries inventory cost, warehouse space and administrative overhead while returning little, and it is the natural candidate for a range rationalisation exercise. The analysis identifies which products would survive one.

5 Limitations and further work

5.1 Limitations

The analysis uses the seven fields specified by the task. The complete file contains seventeen further columns, among them the discount applied, the customer segment and the shipping mode, none of which is used here. Several of the findings above would be explained rather than merely described if those fields were admitted.

The figures reported are descriptive. No statistical test is performed, and no confidence interval accompanies any quantity. The differences discussed are large enough that this is unlikely to change the conclusions, but the point should be stated rather than assumed.

5.2 Further work

1. **Determine whether discount explains the loss-making orders.** The complete dataset contains a discount column. Relating discount to negative profit would convert the second insight from a symptom into a diagnosis, and it is the single most valuable extension available.

2. **Separate seasonality from growth.** The moving average of Figure 2 indicates that both are present. A seasonal decomposition would quantify how much of the December peak is seasonal and how much reflects underlying growth.

3. **Establish whether the loss-making orders concentrate in Furniture and in Southeast Asia.** If the first, second and fourth insights are one problem observed in three ways, a single intervention addresses all three. If they are independent, three are required.

Deliverables accompanying this report

File	Content
`Task1_Sales_Analysis.ipynb`	Complete analysis, executed, with outputs
`report.pdf`	The present report
`INSIGHTS.md`	The five insights, standalone
`README.md`	Guide to the submission
`data/`	Original and cleaned datasets
`charts/`	The five figures, PNG
`results/`	Six tables, CSV, and one Excel workbook

KOUAME Koffi Fidèle
Data Analysis Internship
koffifidelek59@gmail.com

Dataset: Superstore Sales Dataset, 2011 to 2015. Kaggle: `https://www.kaggle.com/datasets/vivek468/superstore-dataset-final`